Supplementary Table 1. Search Query

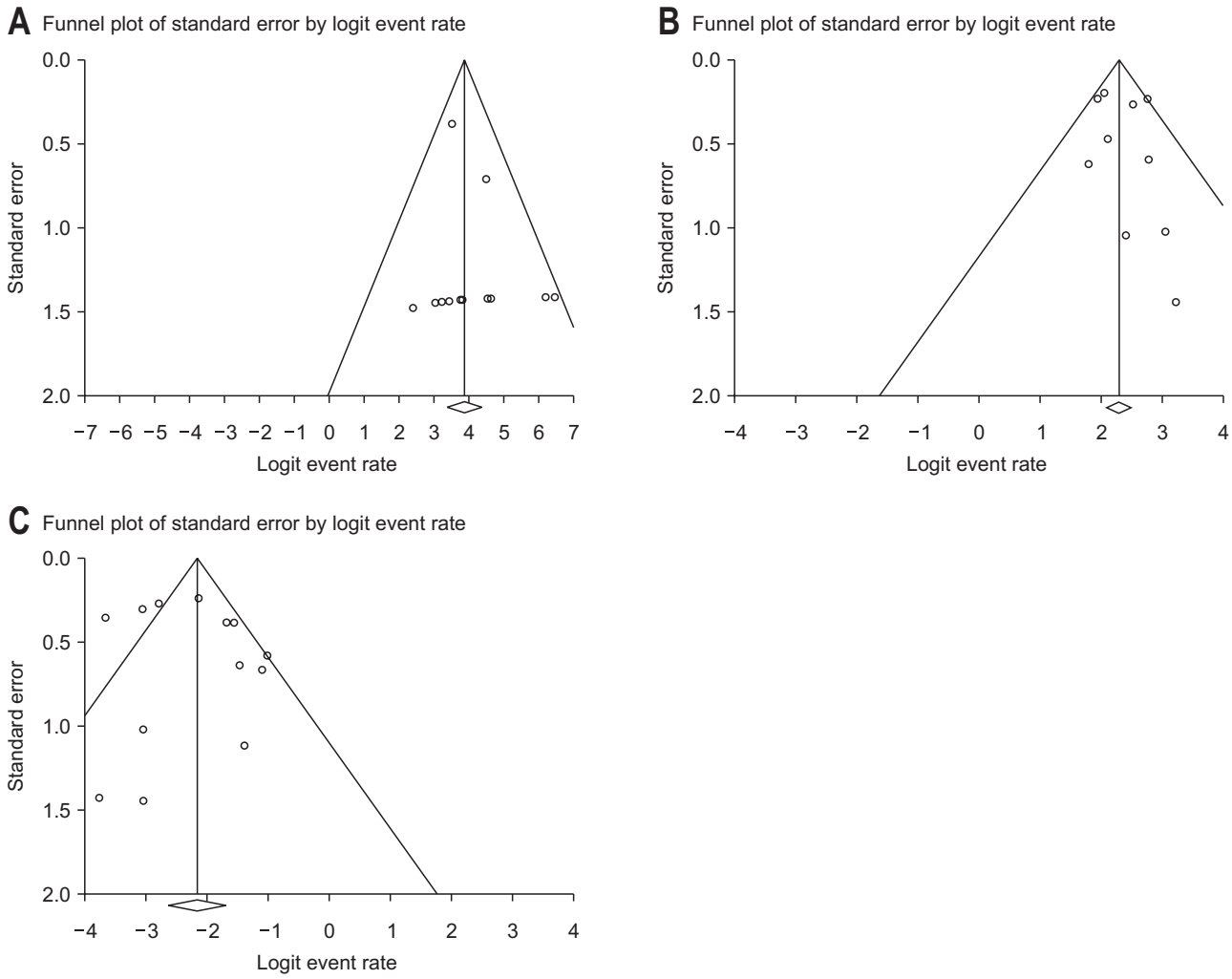
PubMed	Peroral endoscopic myotomy OR per-oral endoscopic myotomy OR POEM OR peroralmyotomy OR "endosopicmyotomy" OR peroralendoscop* OR peroralesophag* OR Heller myotomy [MeSH Terms] OR Myotomy, Heller OR Heller's Myotomy OR Myotomy, Heller's OR Heller Groeneweldt Surgery OR Heller Operation OR Heller Groeneweldt Operation OR Laparoscopic Heller Myotomy OR Heller Myotomy, Laparoscopic OR Laparoscopic Heller Cardiomyotomy OR Pneumatic dilation OR PD OR PBD OR Pneumatic balloon dilation OR Balloon dilation OR BD OR Botox injection OR BI OR Botulinum toxin injection OR BTI AND Esophageal Achalasia [MeSH Terms] OR Achalasias, Esophageal OR Megaesophagus OR Esophageal Achalasias OR Cardiospasm OR Cardiospasms OR Achalasia OR Achalasias OR Achalasia, Esophageal
Embase	"peroral endoscopic myotomy":ab,ti OR "per-oral endoscopic myotomy":ab,ti OR "peroralmyotomy": ab,ti OR "endoscopic myotomy":ab,ti OR peroral NEAR/2 myotom* OR endoscop* NEAR/2 myotom* OR esophag* NEAR/2 myotom* OR "poem":ab,ti AND 'esophageal achalasia' /exp OR achalasia*:ab,ti OR megaesophag*:ab,ti OR cardiospasm*:ab,ti AND [embase]/lim
MEDLINE	"peroral endoscopic myotomy" [tiab] OR "per-oral endoscopic myotomy"[tiab] OR "peroralmyotomy" [tiab] OR"endosopicmyotomy"[tiab] OR peroralendoscop* [tiab] OR peroralesophag*[tiab] OR"POEM"[tiab] AND "Esophageal Achalasia" [MH] OR Achalasia* [tiab] OR megaesophag* [tiab] OR cardiospasm* [tiab]
Cochrane	peroral endoscopic myotomy OR per-oral endoscopic myotomy OR POEM

Supplementary Table 2. Pre- and Post-POEM Eckardt Score and LES Pressure

Author	Group	Eckardt score, mean±SD (range)			LES pressure, mean±SD (range), mm Hg		
		Preoperative	Postoperative	Final FU	Preoperative	Postoperative	Final FU
Tyberg *et al.*[16]	46 PTF	4.3±2.48 (1–10)	NA	1.64±1.67 (0–7) (3-mo FU)	NA	NA	NA
Tyberg *et al.*[21]	51 PTF	7.98±2.11	NA	1.72±1.5 (≥12-mo FU)	NA	NA	NA
Onimaru *et al.*[25]	10 PTF	6.5±1.3	NA	1.1±1.3 (3 mo)	22.1±6.6	NA	10.9±4.5 (3-mo FU)
Vigneswaran *et al.*[26]	5 PTF	6.8 (5–10)	NA	0.6 (148.2 day)	7.5	NA	NA
Zhou *et al.*[28]	12 PTF	9.2±1.1 (7–11)	NA	1.3±1.3 (0–4)	29.4±5.8 (23.5–40.2)	NA	13.5±4.3 (8.6–22.5)
Ling *et al.*[17]	21 PTF 30 Naïve	7.3±2.1 6.8±1.7	0.3±0.5	0.7±0.6 0.5±0.8	31.2±14.2 32.9±13.7	4.2±3.3	7.1±6.8 6.7±5.4
Ngamruengphong *et al.*[18]	90 PTF with HM 90 PTF without HM	7.1±2.7 6.9±2.3		2.12±2.5 (n=85) 1.06±1.2 (n=82) (≥3-mo FU)	25.5±23.2 35.7±32.8 (2 groups total n=90)	NA NA	14.9±10.1 11.3±8.4 (2 groups total n=79)
Tang *et al.*[19]	22 PTF 39 Naïve	7.4±2.4 7.3±1.8	NA NA	1.2±1.1 1.2±1.1 (1-yr FU)	41.1±15.7 39.9±14.4	NA NA	13.4±5.3 (n=12) 14.5±5.6 (n=38)
Kristensen *et al.*[20]	14 PTF with HM	7 (2–11)	NA	4 (1–11) (n=14) (3-mo FU)* 4 (2–7) (n=9) (12-mo FU)* 5 (3–10) (n=7) (24-mo FU)*	13.5 (5–21)*	NA	8 (0–35) (3-mo FU)* 7 (5–16) (12-mo FU)*
	52 PTF without HM	7 (3–12)*	NA	1 (0–8) (n=52) (3-mo FU)* 1 (0–6) (n=40) (12-mo FU)* 2 (0–4) (n=25) (24-mo FU)*	36.5 (11–100)*	NA	13 (2–38) (3-mo FU)* 16 (3–42) (12-mo FU)*
Orenstein *et al.*[23]	16 PTF 24 Naïve	NA NA	NA NA	NA NA	NA NA	NA NA	NA NA
Nabi *et al.*[24]	242 PTF 260 Naïve	7.0±1.4 7.1±1.6	NA NA	1.11±0.74 (n=151) 1.10±0.74 (n=175) (1-yr FU)	36.3±12.1 34.6±13.9	NA NA	13.0±5.02 (n=148) 12.5±4.47 (n=125) (1-yr FU)
Sharata *et al.*[27]	12 PTF 28 Naïve	5 (median) 6 (median)	All<3	1 (median) 1 (median) (6-mo FU)	47.15 40.25	NA NA	NA NA
Zhang *et al.*[22]	46 PTF‡	7 [6, 8], (3–12)*,†	NA	1 [0, 1], (0–4)†	20.7 [16.0, 39.9], (9.3–66.4)*,†	NA	15.6 [12.6, 22.3], (4.7–27.3)*,†
	272 PTF§	8 [7, 9] (4–12)*,†	NA	0 [0, 1], (0–6)†	44.3 [32.4, 56.9], (8.4–119.6)*,†	NA	18.4 [13.1, 25.9], (1.2–50)*,†
Jones *et al.*[29]	15 PTF 30 Naïve	NA NA	NA NA	NA NA	NA NA	NA NA	NA NA
Liu *et al.*[30]	245 PTF 604 Naïve	8 (4–12)* 7 (4–12)*	NA NA	1.69 (0–10)* 1.37 (0–7)*	27 (15–71)* 30 (15–78)*	NA NA	12.2 (6.4–32.2)* 11.1 (4–43.1)* (2 groups total n=438)

POEM, peroral endoscopic myotomy; LES, lower esophageal sphincter; FU, follow-up; PTF, previous treatment failure; NA, not available; HM, Heller myotomy.

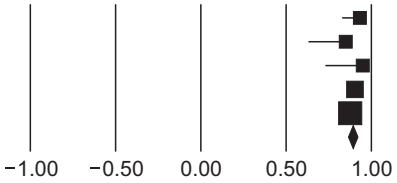
*Median (range); †Median [interquartile range], range; ‡Previous surgical and endoscopic treatment failure; §Previous endoscopic treatment failure.



Supplementary Fig. 1. Funnel plot for the included studies. (A) Technical success; (B) clinical success; (C) adverse events.

A 1 Year follow-up

Study name	Event rate	Lower limit	Upper limit	p-value
Tyberg A *et al.* 2016	0.941	0.833	0.981	0.000
Ling T *et al.* 2014	0.857	0.639	0.953	0.004
Tang X *et al.* 2017	0.955	0.739	0.994	0.003
Nabi Z *et al.* 2018	0.912	0.857	0.947	0.000
Liu ZQ *et al.* 2019	0.886	0.839	0.920	0.000
	0.899	0.869	0.923	0.000



B 2 Years follow-up

Study name	Event rate	Lower limit	Upper limit	p-value
Nabi Z *et al.* 2018	0.842	0.757	0.901	0.000
Liu ZQ *et al.* 2019	0.865	0.817	0.903	0.000
	0.858	0.817	0.891	0.000



C 3 Years follow-up

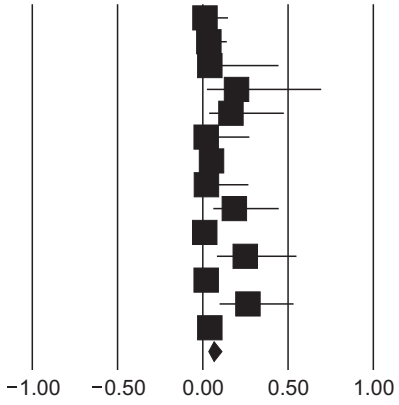
Study name	Event rate	Lower limit	Upper limit	p-value
Nabi Z *et al.* 2018	0.763	0.604	0.872	0.002
Liu ZQ *et al.* 2019	0.820	0.767	0.864	0.000
	0.812	0.762	0.854	0.000



Supplementary Fig. 2. Clinical success of peroral endoscopic myotomy in patients with previous intervention(s). (A) 1-Year follow-up; (B) 2-year follow-up; (3) 3-year follow-up.
CI, confidence interval.

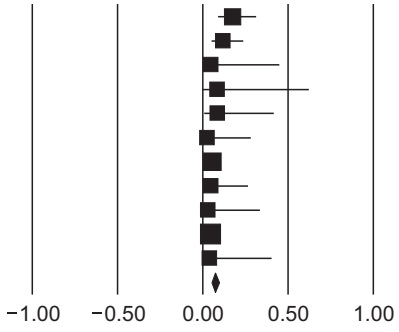
A Major adverse event

Study name	Event rate	Lower limit	Upper limit	p-value
Tyberg A *et al.* 2016	0.011	0.001	0.149	0.001
Tyberg A *et al.* 2018	0.039	0.010	0.144	0.000
Onimaru M *et al.* 2013	0.045	0.003	0.448	0.035
Vigneswaran Y *et al.* 2014	0.200	0.027	0.691	0.215
Zhou PH *et al.* 2013	0.167	0.042	0.477	0.038
Ling T *et al.* 2014	0.023	0.001	0.277	0.009
Ngamruengphong S *et al.* 2017	0.056	0.030	0.100	0.000
Tang X *et al.* 2017	0.022	0.001	0.268	0.008
Orenstein SB *et al.* 2015	0.188	0.062	0.447	0.022
Nabi Z *et al.* 2018	0.012	0.004	0.038	0.000
Sharata A *et al.* 2013	0.250	0.083	0.552	0.099
Zhang X *et al.* 2018	0.025	0.013	0.049	0.000
Jones EL *et al.* 2015	0.267	0.104	0.533	0.083
Liu ZQ *et al.* 2019	0.045	0.025	0.079	0.000
	0.064	0.036	0.111	0.000



B Minor adverse event

Study name	Event rate	Lower limit	Upper limit	p-value
Tyberg A *et al.* 2016	0.174	0.089	0.311	0.000
Tyberg A *et al.* 2018	0.118	0.054	0.238	0.000
Onimaru M *et al.* 2013	0.045	0.003	0.448	0.035
Vigneswaran Y *et al.* 2014	0.083	0.005	0.622	0.105
Zhou PH *et al.* 2013	0.083	0.012	0.413	0.022
Ling T *et al.* 2014	0.023	0.001	0.277	0.009
Ngamruengphong S *et al.* 2017	0.050	0.026	0.093	0.000
Tang X *et al.* 2017	0.045	0.006	0.261	0.003
Orenstein SB *et al.* 2015	0.029	0.002	0.336	0.015
Nabi Z *et al.* 2018	0.045	0.025	0.080	0.000
Sharata A *et al.* 2013	0.038	0.002	0.403	0.026
	0.070	0.052	0.095	0.000

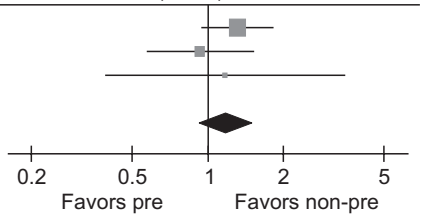


Supplementary Fig. 3. Adverse events of peroral endoscopic myotomy in patients with previous intervention(s). (A) Major adverse events; (B) minor adverse events.
CI, confidence interval.

A Reflux esophagitis at EGD

Study or Subgroup	Pre Events	Pre Total	Non-pre Events	Non-pre Total	Weight	Risk ratio M-H, fixed, 95% CI
Liu ZQ *et al.* 2019	46	202	80	462	60.5%	1.32 [0.95, 1.82]
Nabi Z *et al.* 2018	24	116	29	131	33.8%	0.93 [0.58, 1.51]
Tang X *et al.* 2017	4	17	7	35	5.7%	1.18 [0.40, 3.48]
Total (95% CI)		335		628	100.0%	1.18 [0.91, 1.53]
Total events	74		116			

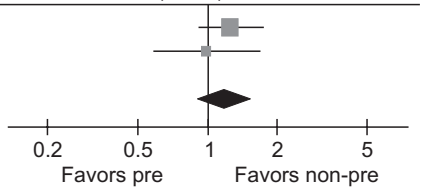
Heterogeneity: Chi^2=1.34, df=2 (p=0.51); I^2=0%
Test for overall effect: Z=1.24 (p=0.21)



B Symptomatic reflux

Study or Subgroup	Pre Events	Pre Total	Non-pre Events	Non-pre Total	Weight	Risk ratio M-H, fixed, 95% CI
Liu ZQ *et al.* 2019	46	245	89	604	69.1%	1.27 [0.92, 1.76]
Nabi Z *et al.* 2018	24	146	22	134	30.9%	1.00 [0.59, 1.70]
Total (95% CI)		391		738	100.0%	1.19 [0.90, 1.57]
Total events	70		111			

Heterogeneity: Chi^2=0.58, df=1 (p=0.45); I^2=0%
Test for overall effect: Z=1.24 (p=0.22)

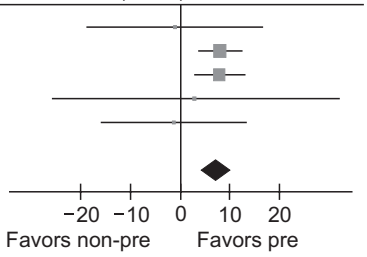


Supplementary Fig. 4. Comparison of the gastroesophageal reflux disease rate between patients with and without previous intervention(s). (A) Reflux esophagitis shown with esophagogastroduodenoscopy (EGD); (B) symptomatic reflux.
M-H, Mantel-Haenszel; CI, confidence interval.

A Procedure time

Study or Subgroup	Pre Mean	Pre SD	Pre Total	Non-pre Mean	Non-pre SD	Non-pre Total	Weight	Mean difference IV, fixed, 95% CI
Jones EL *et al.* 2015	102	29	15	103	27	30	3.3%	−1.00 [−18.57, 16.57]
Ling T *et al.* 2014	42.4	8.3	21	34.3	7.4	30	51.5%	8.10 [3.67, 12.53]
Nabi Z *et al.* 2018	74.9	30.6	242	67	27.1	260	39.2%	7.90 [2.83, 12.97]
Sharata A *et al.* 2013	134	43	12	131	41	28	1.2%	3.00 [−25.68, 31.68]
Tang X *et al.* 2017	60.8	30.9	22	62	21	39	4.8%	−1.20 [−15.70, 13.30]
Total (95% CI)			312			387	100.0%	7.21 [4.04, 10.39]

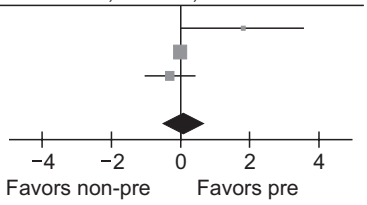
Heterogeneity: Chi^2=2.44, df=4 (p=0.66); I^2=0%
Test for overall effect: Z=4.45 (p<0.001)



B Hospital stay

Study or Subgroup	Pre Mean	Pre SD	Pre Total	Non-pre Mean	Non-pre SD	Non-pre Total	Weight	Mean difference IV, random, 95% CI
Jones EL *et al.* 2015	2.7	3.45	15	0.88	0.25	30	10.4%	1.82 [0.07, 3.57]
Nabi Z *et al.* 2018	3.0	0.54	242	3.0	0.53	260	57.7%	0.00 [−0.09, 0.09]
Tang X *et al.* 2017	6.2	1.3	22	6.5	1.6	39	31.9%	−0.30 [−1.04, 0.44]
Total (95% CI)			279			329	100.0%	0.09 [−0.53, 0.71]

Heterogeneity: Tau^2=0.17; Chi^2=4.80, df=2 (p=0.09); I^2=58%
Test for overall effect: Z=0.29 (p=0.770)



Supplementary Fig. 5. Comparison of the procedure time and length of hospital stay between patients with and without previous intervention(s). (A) Procedure time; (B) hospital stay.
IV, inverse variance; CI, confidence interval.